SCAPE: Search-Based LLM Uncensoring Completed Within Resource-Constrained Environments

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Abstract

Directional ablation, based on the finding that refusal responses in large language models (LLMs) are mediated by a single direction in the residual stream (Arditi et al., 2024), is an established method for removing a model's refusal mechanism without additional training. heretic, a pioneering tool that automates the search process, evaluates each candidate by full-text generation on about 100 prompts; it therefore presupposes a GPU and does not support CPU environments. This paper proposes SCAPE (Structured Cached Abliteration with Prefix Evaluation), a search-based uncensoring algorithm that runs to completion even in a consumer-grade environment without a GPU (CPU only, 16 GB RAM), together with Tema_Q-Abliterated, the tool that implements it. SCAPE has five novel elements. (1) Using as a proxy metric a soft refusal rate defined as a "multi-position vocabulary probability-mass ratio along the base continuation" in place of full-text generation, we formulate uncensoring as a "constrained optimization that preserves quality (KL) as the objective and reduces the refusal rate as a constraint" (candidate evaluation up to about 49× faster than full-text-generation evaluation). (2) We discovered a structural lower bound that the proxy metric retains even under complete uncensoring, the "grammatical floor," and introduced floor calibration, which measures it on a seed grid at the start of the search and recalibrates the constraint value of the objective function. (3) A structured seed grid covering the strength × bandwidth spectrum, including the uniform all-layer orthogonalization of Arditi et al. (the reference solution), removes initialization bias. (4) An elastic prefix cache and an architecture-agnostic swapped partial forward reduce prefix computation to a single pass over the entire search. (5) INT4 weight quantization, streaming loading, and a memory guard enable search on 4B-class models in a CPU-only environment with 16 GB RAM. On Qwen3.5-4B, on the same Google Colab T4 GPU, heretic required over 4 hours and 85–95 trials and reached a best refusal rate of only 27–74%, whereas SCAPE achieved a refusal rate of 8% (KL 0.0266) in 61 minutes and 59 trials. Moreover, in a GPU-less Windows CPU environment (Core i7, 16 GB RAM), it reached a refusal rate of 7% (KL 0.0335) in 6 hours 35 minutes, making search-based uncensoring practical in an environment where heretic cannot run. On a 0.8B model, with 2 CPU cores, it reduced the full-text-generation refusal rate from 0.900 to 0.000 (KL 0.0186) in 148 seconds and 12 trials.

1. Introduction

Instruction-following large language models (LLMs) are equipped with the ability to refuse harmful requests through alignment training. Arditi et al. (2024) [1] showed that this refusal behavior is mediated by a **single direction** (the refusal direction) in the residual stream, and that the refusal mechanism can be removed without additional training by **directional ablation (abliteration)**, which removes the direction —estimated from the mean difference of hidden states between harmful and harmless prompt sets—from the projection matrices of each layer.

In practice, however, which layers to ablate, with what strength, and in which components depends on the model: ablating too strongly destroys general capability, while ablating too weakly leaves refusals in place. A pioneering tool that automates this choice of layers and strength is heretic [2]. heretic performs multi-objective optimization of directional-ablation parameters (simultaneously minimizing the refusal count and the KL divergence) with the TPE-based sampler of Optuna [3], but because it evaluates each candidate by **full-text generation on about 100 prompts**, it presupposes a GPU and does not support CPU environments. In fact, on the freely available Google Colab T4 GPU (session limit 5 h 30 min), it is difficult to run heretic's default configuration (200 trials) to completion on a 4B-class model, and the number of trials must be reduced and the run wrapped up early so that a session disconnection does not lose the computation (§4.3). With one's own GPU there is no time limit, but as §4.3 shows, even on the same GPU the large evaluation cost limits search efficiency. Search-based uncensoring has thus been a technique limited to "users with access to abundant GPU time."

Meanwhile, many users engaged in research and education on the internal refusal mechanisms of models have access only to consumer-grade PCs without GPUs (typically a Windows CPU environment with 16 GB RAM). This paper proposes **SCAPE (Structured Cached Abliteration with Prefix Evaluation)**, an algorithm that realizes **search-based uncensoring that runs to completion in such resource-constrained environments**. SCAPE is implemented in the open-source tool **Tema_Q-Abliterated** (MIT License, published on PyPI [4]), and all experiments in this paper were carried out with that tool.

The contributions of this paper can be summarized in the following six points.

- **C1 (Formulation)**: We formulate uncensoring not as "minimization of the refusal rate" but as a constrained optimization that **maximizes preservation of the original model distribution (KL) subject to the constraint of achieving a target refusal rate**, and replace candidate evaluation by full-text generation with a proxy evaluation based on a "multi-position vocabulary probability-mass ratio along the base continuation" (up to about 49× faster than full-text-generation evaluation, §4.2) (§3.1, §3.2).
- **C2 (Grammatical floor)**: We discovered and measured that the proxy metric has a structural lower bound of 0.3–0.4 (the **grammatical floor**) even under complete uncensoring, and introduced **floor calibration**, which measures the floor at the start of the search and recalibrates the constraint value of the objective function (§3.3). This is a measurement-theoretic framework applicable to constrained optimization with proxy metrics in general.
- **C3 (Seed grid)**: With a structured seed grid that systematically covers the strength × bandwidth spectrum, we include the reference solution of Arditi et al. (uniform all-layer orthogonalization) within about 8 evaluations at the start of the search, eliminating entrapment in local optima (§3.4).
- **C4 (Prefix cache)**: By exploiting the fact that computation shallower than the perturbed layers is invariant across all candidates, a multi-boundary prefix cache and an architecture-agnostic swapped partial forward that reuses the model's own forward as is reduce prefix computation over the whole search to a single pass (§3.7).
- **C5 (Resource saving)**: Combining INT4 weight quantization (memory about 1/3.6 of bf16), streaming loading, and a memory guard, we made search on 4B-class models practical in a CPU-only environment with 16 GB RAM (§3.7, §4.6).
- **C6 (Verification protocol)**: We designed a verification gate based on the floor-free first-token ratio, together with a two-stage evaluation by full-text generation and a human-in-the-loop candidate selection protocol, guaranteeing that the bias of the proxy metric does not remain in the deliverable (§3.8).

2. Related Work

2.1 Directional Ablation

Arditi et al. (2024) [1] proposed estimating the refusal direction v as the mean difference of hidden states at the final position between harmful and harmless prompts, and removing the directional component from the projection matrix W of each layer as $W' = W - v\,v^{T} W$. This single solution, applied uniformly to all layers, is powerful, but it does not take into account differences in per-layer contribution or adjustment of strength. An improvement that additionally uses projection correction (removing the component parallel to the harmless mean direction) is also known in the community.

2.2 Automating the Search: heretic

heretic [2] is a tool that performs multi-objective optimization of directional-ablation parameters (range of perturbed layers, per-component strength, interpolation position of the direction, etc.) with the TPE sampler of

Optuna [3], simultaneously minimizing the refusal count and the KL divergence from the original model. The default configuration is 200 trials (60 random initialization + 140 by TPE), and about 100 prompts' worth of response generation is used to evaluate each candidate. In addition to Dense models, it supports many multimodal models, several MoE architectures, and hybrid models such as Qwen3.5, but it presupposes a GPU and does not support CPU environments. §4.3 of this paper gives a direct comparison on the same model and the same GPU environment.

2.3 Comparison with Prior Methods

Table 1 contrasts the three. The features of SCAPE are (1) drastically reducing search cost by replacing candidate evaluation with a proxy metric, (2) measuring the attainable lower bound of the proxy metric (the grammatical floor) to calibrate the constraint, and (3) treating CPU-only environments as a first-class target.

Table 1 Comparison with prior methods

Aspect	Arditi et al. [1]	heretic [2]	This work (SCAPE)
Scope	Uniform all-layer directional orthogonalization (single solution)	Multi-objective search by Optuna TPE (GPU required)	Constrained objective + floor calibration + seed-grid search
Candidate evaluation	— (confirmed by full-text generation)	Response generation on about 100 prompts	Multi-position mass ratio along the base continuation (up to about 49× faster)
Attainability of target	—	Not stated	**Measures the grammatical floor to calibrate the effective target**
Quality metric	Qualitative observation of outputs	KL divergence from the original model	Multi-position teacher-forced KL + base-ratio NLL + three-layer collapse check
CPU 16 GB environment	Not anticipated	Not supported	Supported via INT4, streaming loading, and memory guard
Verification and adoption	Manual	Full-text generation + manual selection	Floor-free gate + two-stage full-text-generation evaluation + human-in-the-loop selection

3. Proposed Method: SCAPE

SCAPE is a search algorithm implemented in the tool Tema_Q-Abliterated [4]; its processing flow is shown in Fig. 1. The components are described in detail below.

Fig. 1. Processing flow of Tema_Q-Abliterated (with SCAPE)

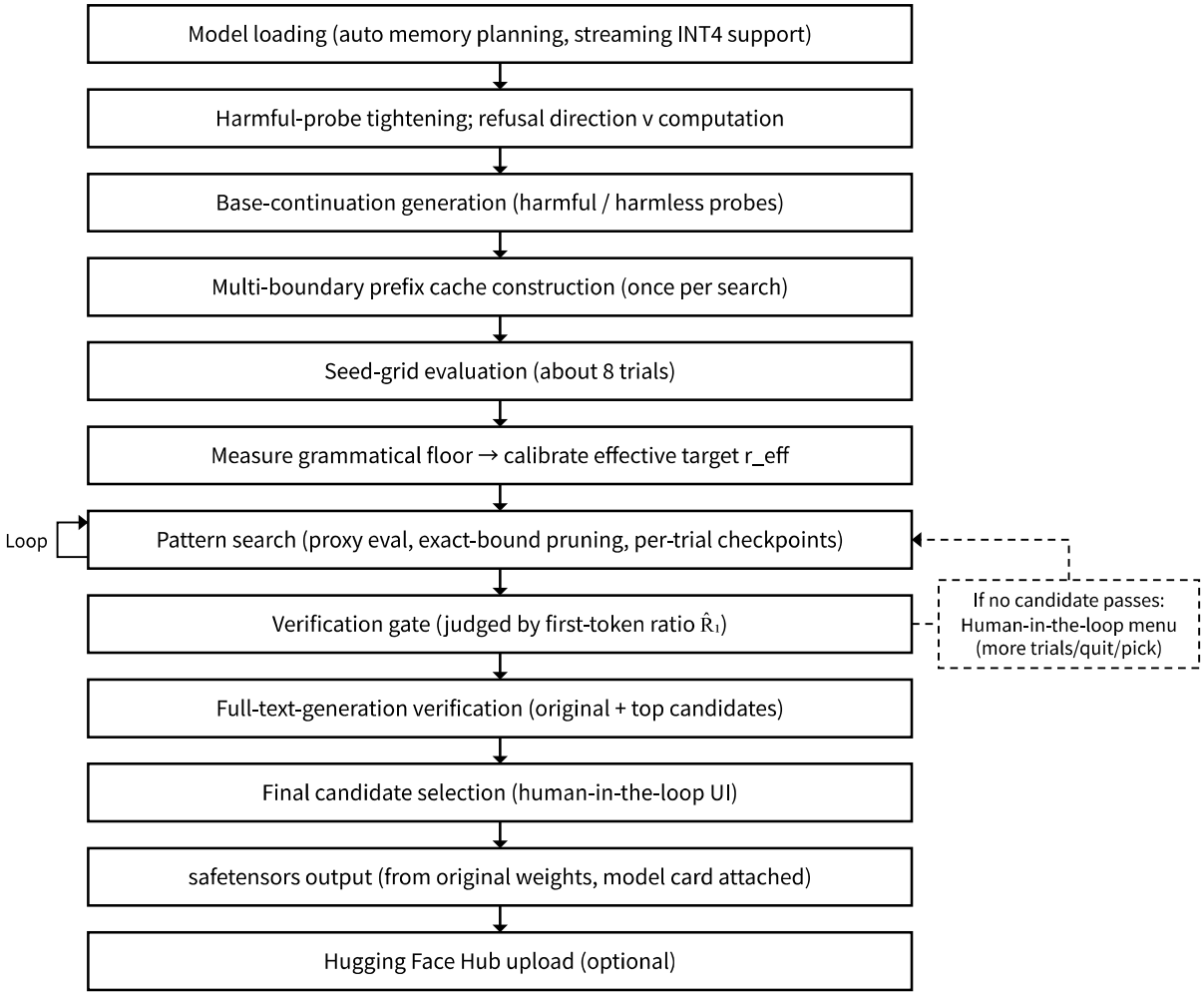


Fig. 1. Processing flow of Tema_Q-Abliterated. From model loading through floor measurement and effective-target calibration with the seed grid, pattern-search exploration, full-text-generation verification after the verification gate, human-in-the-loop candidate selection, and safetensors output, the whole pipeline is automated end to end. If no candidate passes the verification gate, the system moves to the human-in-the-loop menu rather than writing out a model without permission.

3.1 Problem Formulation

The uncensoring operation has a trade-off: if it is too strong, it destroys the model's general capability. SCAPE therefore formulates uncensoring as the following **constrained optimization**.

Eq. (1) Objective function

$$J(\theta) = KL(\theta) + \lambda \cdot \max(0, \hat{R}(\theta) - r_{eff})^2$$

KL(θ): multi-position teacher-forced KL divergence on harmless prompts (quality metric). $\hat{R}(\theta)$: soft refusal rate on harmful probes (proxy metric, Eq. (2)). r_{eff}: effective target refusal rate (calibrated by Eq. (4)). λ: penalty coefficient (default 8.0). θ: parameters of the perturbation kernel family (center, radius, strength, and shape of each component, and the direction mode).

The key point of this formulation is that it does not maximize "how much refusal was removed" but optimizes **how well the original model's distribution is preserved, with achievement of the target refusal rate as a constraint**. Candidates that have reached the target refusal rate compete on the smallness of KL (= preservation of the original model's capability), so quality degradation from excessive perturbation is structurally suppressed.

3.2 Proxy Metric: Mass-Weighted Multi-Position Soft Refusal Rate

Evaluation by full-text generation requires O(number of prompts × generation length) forwards per candidate, which makes a search over tens to hundreds of candidates infeasible in a CPU environment. SCAPE replaces it with the following proxy evaluation. First, at startup, a greedy continuation (K tokens) of each harmful probe is generated with the unperturbed base model (for an aligned model, most of these are refusal sentences). To evaluate a candidate θ, the logits at the last K+1 positions are obtained from one forward pass that teacher-forces the prompt plus the base continuation; at each position the softmax probability masses of the refusal vocabulary ("I", "Sorry", "However", "cannot", etc.) and the compliance vocabulary ("Sure", "Okay", "Here", etc.) are computed, and the mass-weighted ratio is taken across positions.

Eq. (2) Mass-weighted multi-position soft refusal rate

$$\hat{R}(\theta) = \Sigma_t P_t(\text{ref}) \;/\; (\Sigma_t P_t(\text{ref}) + \Sigma_t P_t(\text{cmp}))$$

$P_t(\text{ref})$, $P_t(\text{cmp})$: probability masses of the refusal vocabulary and the compliance vocabulary at position t. Positions where both vocabularies have small mass do not contribute to the ratio, so a noise floor cannot arise in principle. Responses that "begin with a compliant word and turn to refusal midway" are also captured. As a reference value, the first-token ratio $\hat{R}_1 = P_0(\text{ref})/(P_0(\text{ref})+P_0(\text{cmp}))$ at response position 0 only is also recorded (used in the gate of §3.8).

This proxy evaluation is completed with **one forward pass per probe**, and is measured to be **up to about 49× faster** than full-text-generation evaluation at the scale of 100 prompts (§4.2, Table 4). For the quality metric, we use the KL divergence between candidate and base at all continuation positions, with the base continuation of the harmless probes teacher-forced (multi-position teacher-forced KL, Eq. (3)). Unlike a

KL on the first token only, it can capture degradation in which the output "collapses midway through a response."

Eq. (3) Multi-position teacher-forced KL

$$KL(\theta) = \mathrm{mean}_{probe,\ t<K}\ KL\left(p_\theta(\cdot|x,\ y_{<t}) \ \|\ p_{base}(\cdot|x,\ y_{<t}) \right)$$

3.3 Discovery of the Grammatical Floor and Calibration of the Effective Target (Core of This Method)

The center of the measurement-theoretic novelty of this method lies in the discovery of the **grammatical floor** possessed by the proxy metric $\hat{R}$ and in its calibration. Because $\hat{R}$ is measured on a "base continuation that is a refusal sentence," even a model whose refusal circuit has been completely removed retains probability mass on the tokens needed to "grammatically continue a refusal sentence" ("I", "As", "However", "cannot", etc. are included in the refusal vocabulary list). That is, $\hat{R}$ has a structural lower bound of about 0.3–0.4; this is not measurement error but a structural property of the metric (measurements are shown in Fig. 2 and Table 7 of §4.5).

If the existence of the floor is ignored and the target r* (default 0.03) is placed below the floor, the penalty term $\max(0, \hat{R} - r^*)^2$ **saturates** at the floor, and the search can no longer receive a gradient toward true refusal reduction. Moreover, if the candidates to be verified are judged (gated) by a metric that has a floor, the failure that "all candidates are skipped from verification" becomes inevitable for a model whose floor exceeds the threshold.

SCAPE solves this problem by estimating the floor from the measurements of the strongest candidates in the seed grid (§3.4) at the start of the search, and calibrating the effective target.

Eq. (4) Floor measurement and calibration of the effective target

$$\mathrm{floor} = \min\{\ \hat{R}(s)\ :\ s \in \text{seed grid}\ \}$$

$$r_{eff} = \max(r^*,\ \mathrm{floor} + 0.03)$$

After calibration, J of the recorded trials is recomputed with r_{eff} and the best candidate is re-selected (J before calibration is biased toward candidates that are too strong). floor and r_{eff} are saved in the checkpoint, so no re-measurement is needed when resuming after an interruption.

With this calibration, (1) the penalty term becomes meaningful again, (2) after reaching the floor the search correctly shifts to KL minimization, and (3) false triggering of the gate is avoided. **This framework of "measuring the attainable lower bound of the proxy metric and recalibrating the constraint to an attainable value"** is not confined to the problem specific to this method but is applicable to constrained optimization with proxy metrics in general.

3.4 Initialization by a Structured Seed Grid

The search space of directional ablation consists of (perturbed layer band × strength × component), but conventional initialization tends to be biased toward a "narrow band around the saliency peak," and the reference solution of Arditi et al. (uniform all-layer orthogonalization) is far from the initial position. A local search needs several tens of trials to get there, and failures occur in which it is not reached within the time budget. SCAPE evaluates, in about 8 evaluations at the start of the search, a seed grid (Table 2) that systematically covers the whole spectrum.

Table 2 Composition of the seed grid (about 8 evaluations)

Seed	Kernel	Band	Strength	Role
Base	cosine	Narrow peak band	0	Unperturbed reference point
All-band ×3	flat (new)	Uniform over layers 2..L−1	0.7 / 1.4 / 2.1	Strength spectrum (includes the Arditi reference solution); floor measurement
Single component ×2	flat	Uniform over layers 2..L−1	1.4	Separating the contributions of attn / mlp
Peak band ×2	cosine	Narrow peak band	1.0 / 2.0	Local-optimum family (equivalent to conventional initialization)

The seed grid simultaneously achieves (1) floor measurement (C2), (2) global start-point selection that uses the seed with the smallest calibrated J as the search starting point, and (3) elimination of entrapment in narrow-band local optima. The flat kernel is a newly introduced shape that gives uniform strength within the band, and with c = (2+L−1)/2, r = (L−3)/2 it can exactly represent the uniform all-layer orthogonalization of Arditi et al. Layers 0–1 are excluded from the band because the formation of the residual direction is immature and saliency is low there; as a side effect, the first boundary of the prefix cache (§3.7) is always available for all seeds.

3.5 Trust-Region Pattern Search and Exact Lower-Bound Screening

After the seed phase, a pattern search is performed that evaluates ±step candidates for each coordinate, moves if the candidate improves, and otherwise shrinks the step by a factor of 0.6. The coordinate cycle is (strength, center, radius) × each component + direction mode, and if there is no improvement for two cycles, a perturbation restart is performed by applying Gaussian noise around the best solution.

Because the evaluation of $\hat{R}$ proceeds batch by batch, an **exact lower bound** can be computed from partial sums (since $\hat{R} \geq 0$ for the remaining probes).

Eq. (5) Early rejection by exact lower bound

$J_{lb} = \lambda \cdot \max(0, \hat{R}_{lb} - r_{eff})^2 \geq \text{current best } J \quad \Rightarrow \quad \text{reject}$

This rejection is mathematically exact (it cuts only candidates that cannot beat the best), and it can skip the KL evaluation of hopeless candidates (half of the evaluation cost). It is disabled during the seed phase so as not to distort the floor estimate.

The search can be terminated by a specified number of trials, a time budget (; since it proceeds to verification with the best solution at the time the budget is reached, it is guaranteed that "it finishes on budget in any environment"), or an interrupt, and thanks to the per-trial checkpoint it can be stopped and resumed from any point.`--time-budget-min`

3.6 Realizing the Perturbation: Kernel Family, Rank-1 Hook, and Row-Norm Preservation

The layer-wise distribution of perturbation strength is parameterized by a kernel (cosine: the conventional smooth shape; flat: a newly introduced shape that is uniform within the band). For direction estimation, the mean difference of hidden states at each layer boundary at the final position of the harmful and harmless probe sets is normalized, and a projection correction that removes the component parallel to the harmless mean direction is applied. For architectures such as hyper-connections (where the hidden-state width differs between layers), per-component directions are instead extracted from the mean difference of the output activations of the projection modules.

Since $W' = W - w \cdot v(v^T W)$ is rank-1, the same perturbation can be applied by an output hook **without rewriting the weight matrices**.

Eq. (6) Rank-1 hook application

$u = W^T v$ (precomputed) $\qquad y' = Wx - w \cdot (u^T x) \cdot v$

Because applying and resetting reduce to attaching and detaching hooks, the weight rewriting of several seconds per trial and the GB-scale memory churn disappear, and the same perturbation can also be applied to INT4-quantized modules. At startup the output difference between the two methods is measured and verified, and if it is unacceptable the system automatically falls back to the weight-rewriting (exact) mode. The final output is always computed as a full patch in fp32 from the original safetensors, so the weight precision of the output model does not degrade at all even when quantization is used for the search. The shrinkage of row norms caused by orthogonalization is offset by a row-norm-preserving correction.

3.7 Resource-Saving Mechanisms

Multi-boundary prefix cache: Because the perturbation is confined to the layer band [a, b], the computation below layer a is invariant across all candidates. Before the search begins, all probes are swept once and the activations at multiple layer boundaries (2 to

12 boundaries within the memory budget) are saved. Each subsequent evaluation resumes from the deepest boundary at or below the candidate's perturbation start layer. The cache construction cost is equivalent to one or two ordinary evaluations, and rebuilding occurs only once over the whole search (verified by measurement in §4.4).

Swapped partial forward: Cache injection is realized by "replacing the forward of layers [0..a) with a pass-through, injecting the cached activations at the boundary, and running the model's own forward as is." Because mask construction, RoPE, and the recurrent state of linear attention all go through the model's original code path, no architecture-specific implementation is needed, and it can be applied after numerical equivalence has been verified by measurement, without distinguishing Llama / Qwen / Gemma / hybrid linear attention (on a real Qwen3.5-0.8B, the logits difference at any bypass depth is 0.00e+00, i.e., exact numerical agreement). For structural constraints such as shared KV, multiple bypass depths are measured at startup and the safe maximum depth is determined automatically.

INT4 weight quantization (WQ4G) and memory guard: The perturbation targets and lm_head are left in bf16, and the other Linear weights are packed into group-symmetric INT4 (memory about 1/3.6 of bf16). Execution uses split-nibble matrix multiplication and an LRU elastic dequantization cache, and on a cache hit the speed is identical to bf16. Streaming loading, which packs while reading safetensors in row-chunk units, removes the need for bulk loading, and during execution a memory guard monitors RAM, VRAM, and SWAP and, when conditions become dangerous, performs staged cache shrinking, GC, batch reduction, and automatic retry.

3.8 Verification Gate and Human-in-the-Loop Protocol

The proxy metric is only a means of making the search efficient; adoption of the final deliverable is decided by full-text-generation measurement (two-stage evaluation). The gate for suppressing verification cost (excluding candidates that are "not worth verifying by full-text generation") must be **judged by the floor-free first-token ratio $\mathbf{R\hat{}_1}$** (same rationale as §3.3: $R\hat{}_1$ has no floor because response position 0 is not subject to grammatical constraints, and in the measurements of §4.5 it converges monotonically to 0.02–0.05). The number of top candidates to verify can be changed in the settings (by default, the original model + the top 2 candidates). Furthermore, when no candidate passes the gate, or when the final candidate is adopted, writing out the model is not left to automatic judgment. In interactive environments, a human-in-the-loop menu is presented that lets the user choose to add trials, quit, or verify and write out a specified trial; automatic write-out is performed only when explicitly specified (). This is an operational design that "does not let the results of long computations be discarded without permission, nor adopted without permission," and it is applicable to exploratory systems in general.--`select auto`

4. Experiments

4.1 Experimental Environment

The target model for the comparison experiments is Qwen3.5-4B (a 4B-class instruction-following model with a hybrid linear-attention architecture), and the final

metric common to both tools is the refusal rate (detected by refusal markers) from full-text generation on 100 harmful evaluation prompts. The execution environments are shown in Table 3. Because heretic does not support CPU, it was run only on a Google Colab T4 GPU (session limit 5 h 30 min). Tema_Q-Abliterated was run on the same Colab T4 GPU and also in a consumer-grade environment without a GPU (13th Gen Intel Core i7, 16 GB DRAM, Windows, CPU only). All SCAPE-side experiments are execution results of Tema_Q-Abliterated, and the numbers in the tables and figures are based on the execution records (JSON and PNG reports) output by that tool.

Table 3 Experimental environments

Tool	Environment	Notes
heretic [2]	Google Colab T4 GPU (16 GB VRAM)	CPU not supported. Free session limit 5 h 30 min
Tema_Q-Abliterated	Google Colab T4 GPU (16 GB VRAM)	bf16 evaluation
Tema_Q-Abliterated	13th Gen Core i7, 16 GB DRAM, Windows (no GPU)	INT4 evaluation (streaming loading and memory guard enabled)

4.2 Speed of the Proxy Evaluation (Comparison with Full-Text-Generation Evaluation)

To quantify the speed of the proxy evaluation (§3.2), we measured the evaluation time per candidate in the same environment (Qwen3.5-0.8B, INT4, 2 CPU cores) (Table 4). The proxy evaluation takes 4–7.5 s for one forward pass over 16 probes (simultaneous computation of the multi-position mass ratio $R\hat{}$ and the multi-position KL), whereas the full-text-generation evaluation took about 50 s for 24 prompts × 32 tokens. Scaled to the 100-prompt scale used for verification, this becomes about 208 s, giving a speed ratio of about 28–49× (up to about 49×). This difference is the foundation that makes a search over tens to hundreds of candidates feasible in practical time in a CPU environment.

Table 4 Speed comparison of proxy evaluation and full-text-generation evaluation (Qwen3.5-0.8B, INT4, 2 CPU cores)

Evaluation method	Content	Time per candidate	Relative speed
Proxy evaluation (this method)	16 probes × 1 forward ($R\hat{}$ + multi-position KL)	**4–7.5 s**	1×
Full-text-generation evaluation	24 prompts × 32 tokens (measured)	about 50 s	about 7–12× slower
Full-text-generation evaluation	100 prompts × 32 tokens (converted from measured value)	about 208 s	about 28–49× slower

4.3 Comparison with the Existing Tool heretic (Qwen3.5-4B)

Table 5 shows the results. heretic's default configuration is 200 trials (60 random initialization + 140 by TPE), but considering the session limit of the Colab free tier (5 h 30 min), it is difficult to run the default number of trials to the end on a 4B-class model. So that results midway through the search would not be lost to a session disconnection, we performed two independent runs in a configuration where the number of trials was reduced and the run was wrapped up early (run 1: 85 trials = 20 random + 65 search; run 2: 95 trials = 5 random + 90 search). Even so, the total run times reached 4 h 2 min and 4 h 25 min, consuming 73–80% of the session limit. The full-text-generation refusal rate of the best candidates remained at 74/100 and 27/100. Note that with one's own GPU there is no such time limit and the default number of trials could be run, but as follows, in a comparison on the same GPU, evaluation cost limits both the number of trials and the total time, so the efficiency advantage of this method holds regardless of whether a time limit exists.

Table 5 Comparison with heretic on Qwen3.5-4B (best candidate, full-text generation on 100 prompts, heretic's prompts used for comparison)

	heretic run 1 (Colab T4)	**heretic run 2 (Colab T4)**	**SCAPE (Colab T4)**	**SCAPE (Windows CPU, 16 GB)**
Total run time	4 h 2 min	4 h 25 min	**1 h 1 min**	6 h 35 min
of which search time	3 h 35 min	3 h 25 min	**40 min**	3 h 0 min
Trials (of which random initialization)	85 (20)	95 (5)	59 (0)	21 (0)
Refusal rate of best candidate	74/100	27/100	**8/100**	**7/100**
KL of best candidate	0.0164	0.0391	0.0266	0.0335

By contrast, SCAPE **shortened the total run time to about one quarter (61 min) on the same T4 GPU while achieving a refusal rate of 8/100**, uncensoring far better than both heretic runs. Considering search time alone, it is 40 min versus 3 h 25 min–3 h 35 min, about a 5× speedup. This is the combined effect of (1) the drastic reduction in per-trial cost through proxy evaluation (§4.2), (2) direct arrival near the reference solution through the seed grid (random initialization became unnecessary), and (3) early convergence of the search through floor calibration. Note that the KL computation method is not necessarily identical between the two tools, so this is not a strict equivalence comparison, but both are below 0.04, showing that the original model's distribution is well preserved even though the refusal rate was greatly reduced.

More importantly, SCAPE **runs to completion even in a GPU-less Windows CPU environment (Core i7, 16 GB RAM)**. Although the wall-clock time of 6 h 35 min is longer than on the GPU, in an **environment where heretic cannot run at all** it reached a refusal rate of 7/100 (KL 0.0335), a result better than both heretic runs. This means that

search-based uncensoring changes from "a technique for users with GPU access" to "a technique for general research and education users."

4.4 End-to-End Evaluation and Ablation on a Small Model (Qwen3.5-0.8B)

To verify in detail the accuracy reached in a CPU environment, we performed end-to-end evaluation on Qwen3.5-0.8B (INT4, 2 CPU cores) (Table 6). As the comparison condition, we set up an ablation condition (corresponding to a conventional proxy search) that removes the seed grid and floor calibration from this method and searches with peak-narrow-band initialization and an uncalibrated target value.

Table 6 End-to-end results and ablation on Qwen3.5-0.8B, INT4, 2 CPU cores

Item	Ablation (no seed grid, no floor calibration)	SCAPE (proposed method)
Full-text-generation refusal rate (HF dataset test split, 40 prompts)	0.94 → 0.10 (36 trials, 395 s)	**0.900 → 0.000 (12 trials, 148 s)**
KL of best candidate	0.0237	**0.0186**
Floor of the proxy metric	Not measured (implicit)	Measured 0.316 → calibrated to effective target 0.346
Reaching the reference solution (all-band orthogonalization)	Not reached even in 31 trials	**Reached at the 2nd seed evaluation**
Per evaluation	—	4–7.5 s (16 probes, prefix cache)
Prefix cache rebuild	—	Once over the whole search

Under the ablation condition, a refusal rate of 0.10 remained even after spending 36 trials and 395 s, and the reference solution was not reached even in 31 trials. In contrast, SCAPE reached the reference solution already at the second evaluation of the seed grid (flat 0.7) and reached a full-text-generation refusal rate of 0.000 (KL 0.0186) in 12 trials and 148 s. In the per-trial progression: base 0.598 → 0.410 at all-band 0.7 → floor 0.316 measured at all-band 2.1 → effective target calibrated to 0.346 → thereafter shifting to KL minimization under floor attainment, obtaining $\hat{R}$ = 0.351, $\hat{R}_1$ = 0.016, and KL 0.0186 in the best trial. Full-text-generation verification was only 3 runs in total (the original model + the top 2 candidates), so the bias of the proxy metric does not remain in the deliverable.

4.5 Measurement of the Grammatical Floor and Tracking by the Proxy Metric

To directly verify the existence of the grammatical floor (§3.3), we swept the strength of the uniform all-layer flat kernel over 0 / 0.7 / 1.4 / 2.1 and simultaneously measured the proxy metrics and the full-text-generation refusal rate (ground truth) at each point (Qwen3.5-0.8B, INT4, CPU, 24 probes, full-text generation on 24 prompts × 32 tokens). The results are shown in Fig. 2 and Table 7.

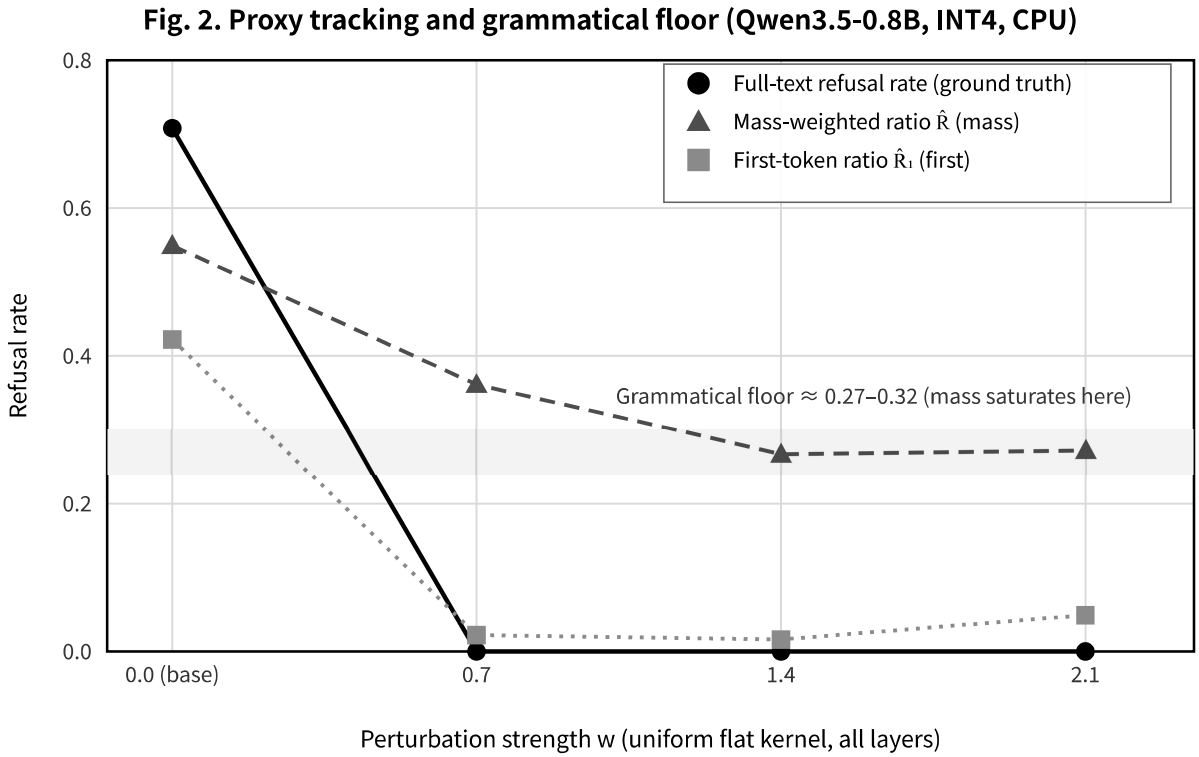


Fig. 2. Proxy-metric tracking and the grammatical floor. Even after the full-text-generation refusal rate (ground truth, circle markers) already reaches 0 at strength 0.7, the mass-weighted multi-position ratio $\hat{R}$ (mass, triangle markers) saturates at 0.267–0.272. This is the grammatical floor. The first-token ratio $\hat{R}_1$ (first, square markers) has no floor and converges to 0.02–0.05, so it is used for the verification gate.

Table 7 Numerical values of the strength-sweep experiment (Qwen3.5-0.8B, INT4, CPU)

Strength w	$\hat{R}$ (mass)	$\hat{R}_1$ (first)	Full-text-generation refusal rate	KL
0.0 (base)	0.549	0.422	0.708	0.000
0.7	0.361	0.022	0.000	0.037
1.4	0.267	0.016	0.000	0.090
2.1	0.272	0.049	0.000	0.173

The mass-weighted ratio $\hat{R}$ moves monotonically with the full-text-generation refusal rate (rank correlation 0.945) and is effective as a metric for directing the search, but its absolute value saturates at the floor. $\hat{R}_1$, on the other hand, converges to near 0 without a floor. This division of roles—"$\hat{R}$ for ordering, $\hat{R}_1$ for absolute-value judgment (the gate)"—is the measurement design of SCAPE. Note that all-band orthogonalization at strength 0.7 alone brings the full-text-generation refusal rate from 0.708 to 0.000 (KL 0.037), confirming the power of the reference solution and the significance of evaluating it directly with the seed grid.

4.6 Convergence Behavior and Memory Behavior in a Windows CPU 16 GB Environment

Fig. 3 shows the convergence of the refusal rate in the Qwen3.5-4B search (21 trials) in the Windows CPU 16 GB environment. After

reaching the best proxy value of 0.34 at trial 4 through the seed-grid evaluation, the search shifted to KL minimization under the calibrated objective function, and trials 14 and 16 went through full-text-generation verification (both with a refusal rate of 7/100) and were adopted as the final model (refusal rate 0.07, original model 0.97). Because the seed grid secures the neighborhood of the reference solution at the start, the practical range is reached with few trials.

Fig. 3. Refusal-rate convergence during search (Qwen3.5-4B, Windows CPU 16GB)

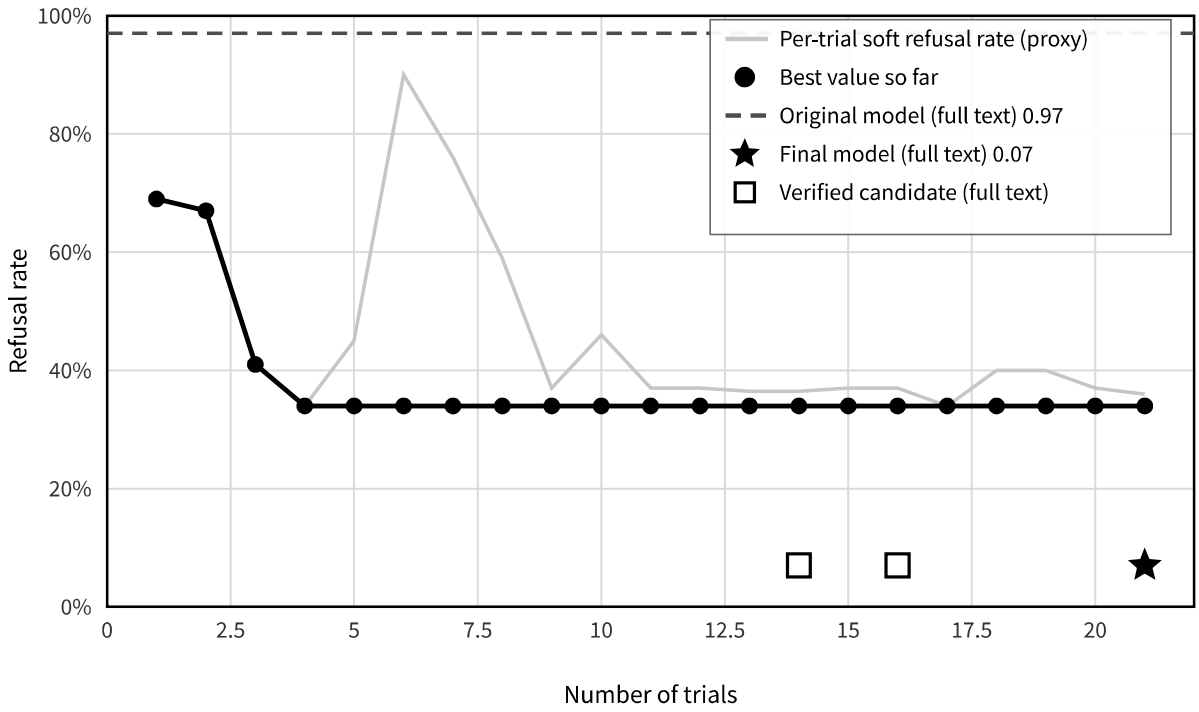


Fig. 3. Search process in the Windows CPU 16 GB environment (Core i7, 16 GB DRAM). The light thin line shows the soft refusal rate (proxy metric) of each trial, and the solid line with circle markers shows the best value so far. The dashed line shows the full-text-generation refusal rate of the original model (0.97), the star shows the adopted final model (0.07), and the open squares show candidates verified by full-text generation.

Fig. 4 shows the memory trajectory of the same run (total run time 6 h 35 min). Because the 4B-class model is handled with INT4 streaming loading, the process RSS is kept to 5.0–5.6 GB and operates stably within the 16 GB RAM constraint. While free RAM declines gradually from 2.9 GB to 2.2 GB, SWAP stays almost constant at about 6.9 GB, showing that the memory guard's monitoring of SWAP increments and the elastic control of the cache are functioning. Since the weights of the output model are generated from the original safetensors rather than from the quantized representation used for the search, this memory saving does not impair the accuracy of the deliverable.

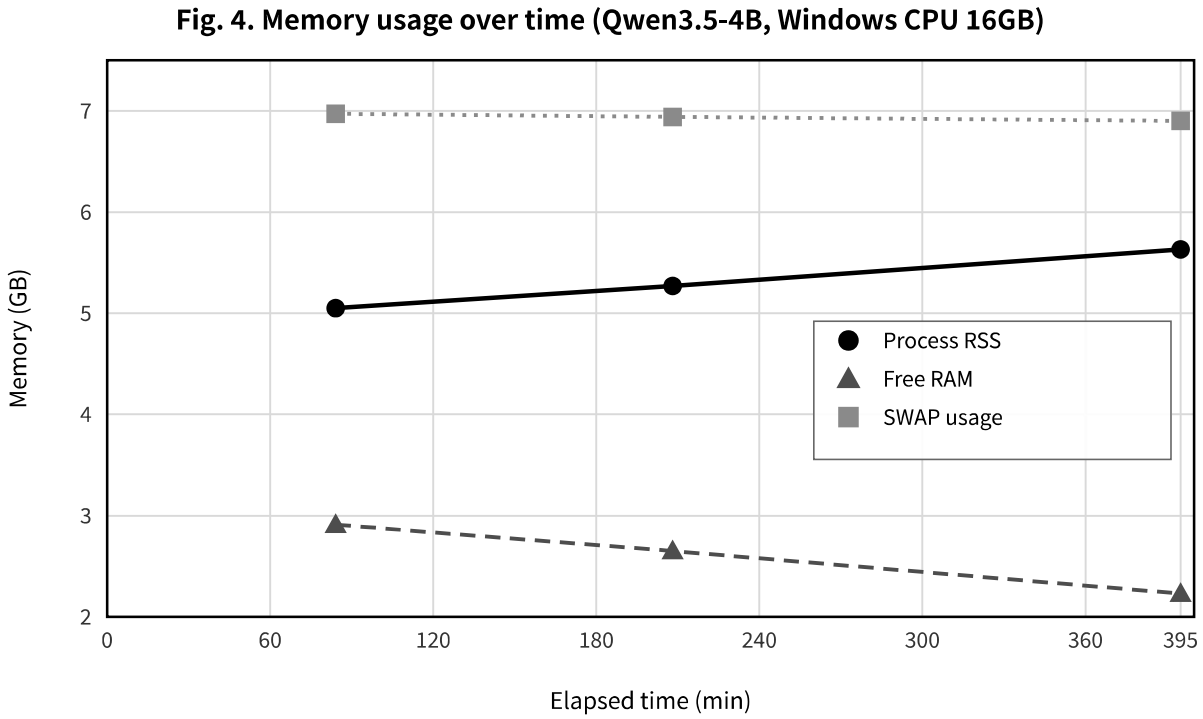


Fig. 4. Memory usage over time in the Windows CPU 16 GB environment. INT4 streaming loading and the elastic cache keep RSS at around 5 GB while holding SWAP almost constant. The circle, triangle, and square markers indicate process RSS, free RAM, and SWAP usage, respectively.

4.7 Architecture Adaptability

Through automatic detection by attribute probing, a single code path handles Dense (Llama / Mistral / Qwen / Gemma / Phi families), MoE (both per-expert Linear and fused 3D expert tensor formats), hybrid linear attention, multimodal (the language-model part of VLMs), hyper-connections (where the residual width differs between layers so a residual direction cannot be defined; per-component directions are extracted from the output activations of the projection modules), and shared KV (the bypass depth is measured at startup and the safe maximum depth is determined automatically). All 8 automated tests with synthetic tiny models (fused-MoE detection, INT4 evaluation, VLM resolution, safe-depth detection for shared KV, streaming loading, export verification, etc.) PASS, and exact agreement of the numerical equivalence of the partial forward on real hardware (logits difference 0.00e+00) has been confirmed.

4.8 Uncensoring on Each Model

Table 8 shows the execution results on the models used in verification.

Table 8 Tool execution results on each model (best candidate, full-text generation on 100 prompts, heretic's prompts used for comparison)

	Qwen3.5-0.8B	Qwen3.5-4B	gemma4-E2B	LFM2.5-2.6B
Trials	12	59	200	85
Refusal rate of best candidate	0/100	8/100	20/100	2/100
KL of best candidate	0.0186	0.0266	0.0199	0.0143

5. Discussion

5.1 Positioning of the Novelty

The core novelty of this method is the **discovery of the grammatical floor and floor calibration**. In constrained optimization with a proxy metric, if the metric has an essentially unreachable lower bound, merely placing the constraint below that bound saturates the objective function and renders the search dysfunctional. The procedure that SCAPE demonstrates —"measure the attainable lower bound with a small structured grid at the start of the search, recalibrate the constraint, and then optimize"—is general enough to apply not only to directional ablation but to constrained optimization driven by proxy metrics at large. In addition, the combination of rank-1 hook application, the multi-boundary prefix cache, and the architecture-agnostic partial forward reduces candidate evaluation cost to "one forward pass from the perturbed layer onward," an engineering contribution that brings search-based weight manipulation into CPU environments for the first time.

5.2 Practicality in Resource-Constrained Environments

As §4.3 shows, SCAPE achieves a lower refusal rate than heretic in about one quarter of the time and with fewer trials on the same GPU. With one's own GPU environment there is no session time limit as on Colab, but low evaluation cost benefits both the number of trials and the total time, so the efficiency advantage of this method holds regardless of whether a time limit exists. Furthermore, it runs to completion even in a Windows CPU environment with no GPU and 16 GB RAM, where heretic cannot run. In practical terms, (1) dependencies are kept to a minimal set such as torch / transformers / safetensors and it can be installed with , (2) specifying a time budget makes it "finish on budget in any environment," (3) it can be interrupted and resumed through per-trial checkpoints, and (4) output is unified as safetensors with direct upload to the Hugging Face Hub and automatic model-card generation, all of which support reproducibility and ease of distribution for research and education. Processing time scales roughly linearly with model size; for example, in a CPU environment with 12–16 threads the guideline is 6–12 min for 0.5–1B-class models, 40–60 min for 4B-class models, and 2–3 hours for 7–9B-class models (INT4), and on a Colab T4 it is 3–6 min, 25–35 min, and 1–1.5 hours, respectively.`pip install temaq-abliterated`

5.3 Limitations

First, because the floor estimate is based on the minimum reached by the seed grid, if the attainable lower bound of the whole search family is even lower, the effective target will

be conservative (likewise when the full-text-generation refusal rate of the strongest seed is not 0 for a robustly aligned model). Even in this case, however, the meaning of the objective function—"maximize quality within the attainable range"—is preserved. Second, KL measurement is limited to the base continuation (K tokens) on harmless probes, so capability degradation in long-form generation must be checked by final verification and human evaluation. Third, pattern search with perturbation restarts does not guarantee a global optimum.

5.4 Ethical Considerations

This tool is designed for research and educational purposes concerning the internal refusal mechanisms of models. Because an uncensored model operates with its safety filters removed, great care is required in handling it, and the original license and terms of use of each model must be observed. The quality metrics of the output model (KL, refusal rate, collapse check) are reference values, and actual behavior varies with prompts, model scale, and seeds.

6. Conclusion

This paper proposed SCAPE, a search-based LLM uncensoring algorithm that runs to completion in resource-constrained environments, and Tema_Q-Abliterated, the tool that implements it. Through the discovery of the structural lower bound of the proxy metric (the "grammatical floor") and floor calibration, a structured seed grid that includes the reference solution, and resource saving by the prefix cache and INT4 quantization, it achieved a lower refusal rate than heretic in about one quarter of the time on the same GPU, and further made uncensoring of a 4B-class model practical in a CPU environment with no GPU and 16 GB RAM, which heretic does not support. Future work includes making the floor estimation more robust, verification on larger models, and extending the search space (global search including discrete combinations of layer selection).

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